

Nikhil Singh

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SKILLS

Languages: Rust, Nix, Go, Java, Python, C/C++, Haskell, Bash, JavaScript/TypeScript

Systems: Linux (NixOS), low-latency systems, debugging, performance optimization

Infrastructure: NixOS/Flakes, Docker, Kubernetes, AWS, GCP, CI/CD (Jenkins, GitHub Actions)

Developer Tools: Git, Neovim, Tmux, QEMU, SSH

Backend Frameworks: Node.js, Django, FastAPI, Flask, Actix Web (Rust), Servant (Haskell)

Datastores & Messaging: MySQL, MongoDB, Redis, Kafka

Engineering Focus: Backend systems, distributed workflows, reproducible infrastructure, performance-sensitive software

EXPERIENCE

Product Engineer | *Juspay Technologies*

Jun 2025 – Present

- Led AI-First Development initiative: deployed company-hosted fine-tuned models via Nix for AI-assisted PR reviews, reducing manual review effort by 80%
- Measured AI performance using LiteLLM to track usage, cost, and security; ensured company IP protection while maintaining developer-friendly access
- Built Project Bernoulli (Rust) handling millions of TPS for enterprise merchants with SDK-based S2S session management
- Wrote custom GPG library in Rust with MDC payload support for secure cryptographic operations in payment flows
- Deployed Nix-based devbox cutting developer onboarding from hours to minutes for 800+ engineers; eliminated environment drift and "works on my machine" issues
- Built event-driven payment services processing millions of UPI transactions daily with sub-100ms latency and 99.99% uptime
- Cut API response times by 60% (200ms to 80ms) using Redis caching and async processing in high-concurrency flows

M.Tech Researcher – Systems Engineering | *Gautam Buddha University*

May 2023 – May 2025

- Authored IEEE paper on modular NixOS configurations for reproducible OS management (IEEE Xplore).
- Co-authored IEEE paper on ML-based heavy rainfall prediction for the Northern Himalayan region (IEEE Xplore).
- Developed Nix flake-based configurations using disko and home-manager for hardware-agnostic deployments.

Software Developer Intern | *Decet Technology*

2023 – 2024

- Initiated and built deployment infrastructure in Go to streamline team workflows; reduced deployment time from hours to minutes (personal initiative leveraging Linux/DevOps expertise)
- Built core Python backend services handling 10k+ daily requests with REST APIs and database integrations

Package Maintainer – Arch Linux & NixOS | *Arch Linux & NixOS Communities*

2020 – 2025

- Built and maintained 15+ AUR/Nix packages with 50k+ downloads; upstreamed 10+ fixes to NixOS (nixpkgs) and Hyprwm; hosted custom binary cache

PROJECTS

Ndots | *NixOS System Configuration*

2024 – Present

- Modular NixOS flake used by 50+ developers for reproducible workstations; supports 5+ machine architectures
- Automated deployment pipeline reducing system setup time from days to under 30 minutes with full rollback capability

Nvix | *Neovim Distribution for Terminal Users*

2024 – Present

- High-performance Neovim configuration optimized for keyboard-centric developers; achieves 50ms startup with 100+ plugins
- Modular plugin architecture supporting LSP, DAP, and Treesitter for 20+ languages with lazy-loading

Tmux Minimal Status | *Terminal UI, Open Source*

2020 – Present

- Maintainer of popular tmux status bar (240+ GitHub stars); optimized for performance and minimal resource footprint in high-frequency terminal environments

Android CI/CD Pipeline | *AOSP, Jenkins, Kernel Development*

2019 – 2022

- Automated Android ROM builds for 15+ devices using Jenkins; delivered OTA updates to 10k+ users with 99% stability
- Debugged kernel panics and hardware abstraction layer issues; patched Qualcomm drivers for custom ROM compatibility

QKM | *QEMU/KVM, VFIO, Low-latency Hardware Passthrough*

2021 – 2023

- Built custom keyboard firmware passthrough achieving 41ms latency using QEMU/KVM with VFIO and PCI device isolation
- Optimized IOMMU group configurations enabling real-time peripheral virtualization for competitive gaming use cases

EDUCATION

Gautam Buddha University

Greater Noida, UP

Master of Technology in Software Engineering (**CGPA: 8.83**)

Aug 2023 – May 2025

Bachelor of Technology in Computer Science Engineering

Aug 2020 – Aug 2023